Field Guide to Alpine Botany

1 Introduction

Alpine plants live in a harsh world of wind, cold and intense ultraviolet light. This guide describes the species most often met by hikers between the forest limit and the permanent snow, and explains how each of them survives the short mountain summer.

2 Plant communities

2.1 Limestone slopes

2.1.1 Edelweiss

Edelweiss grows on limestone above 1800 metres, in sunny and well drained places. Its woolly white bracts protect the tiny flowers from drying winds and from radiation, and the plant is protected by law in most alpine countries.

2.1.2 Alpine gentian

The alpine gentian forms low cushions covered with deep blue flowers in early summer. It prefers short grass on calcareous soil and flowers a few days after the snow has melted.

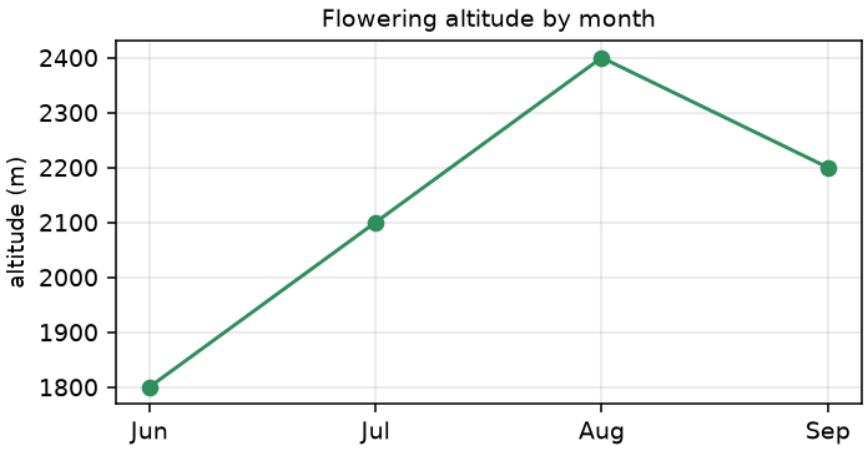

Figure 1. Mean altitude of flowering gentians through the summer.

2.2 Siliceous screes

On acidic rock the vegetation is dominated by cushion plants such as moss campion. Their dense shape traps heat and moisture and lets them colonise ground that moves slowly with frost and thaw.

3 Observation tips

3.1 Best season

The best period for observation runs from late June to the middle of August, depending on altitude and on the date of snow melt. Early morning light is ideal for photography and for counting flowers.

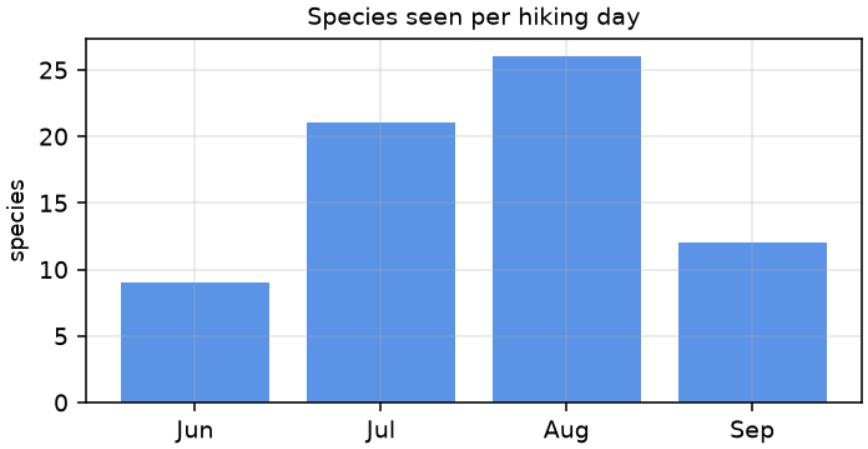

Figure 2. Average number of species recorded per hiking day.

3.2 Respecting the habitat

Stay on marked paths, never pick protected species and keep dogs on a lead. Trampled cushions need decades to recover, so a few careless steps can undo the growth of a whole generation of plants.